Artificial Intelligence-Enabled Digital Twin for Intelligent Building Monitoring and Predictive Analytics

1st DAKSHANAMURTHI S
Artificial Intelligence and Data Science
S.A. Engineering College
Chennai, India
2324020@saec.ac.in

2nd DHANUSH A
Artificial Intelligence and Data Science
S.A. Engineering College
Chennai, India
2324023@saec.ac.in

3rd HEMANTH KUMAR M P
Artificial Intelligence and Data Science
S.A. Engineering College
Chennai, India
2324032@saec.ac.in

4th Muthu Kumar G
Artificial Intelligence and Data Science
S.A. Engineering College
Chennai, India
muthukumarg79.cse@gmail.com

Abstract*— Constant monitoring of environmental conditions and occupant activities in various spaces is essential for effective building management. Traditional monitoring systems can collect sensor measurements but have limitations in organizing the data for analysis and visualization in a timely manner. This paper presents an AI-enabled Digital Twin system that integrates the Internet of Things (IoT), machine learning, and rule-based analysis with real-time data processing and Digital Twin visualization for intelligent building monitoring. The system is built on an Espressif Systems ESP32 microcontroller and is equipped with DHT22 and MQ-2 sensors, as well as PIR motion sensors, to capture environmental conditions in three spaces of the prototype building: Reception, Manager Cabin, and Workspace. The measured data are transmitted to a FastAPI backend via Wi-Fi for storage in a MongoDB database. For analysis, the data are preprocessed, and a machine learning model based on Random Forest is developed to predict room occupancy, while rule-based analysis classifies environmental conditions into normal, warning, and critical states. PIR sensors also provide occupancy-related information. A Digital Twin dashboard is developed using React and provides occupancy-related information along with sensor measurements, historical data, alerts, and condition status of the monitored spaces. The integrated prototype establishes a basis for building automation and monitoring applications and demonstrates the integration of various technologies, including IoT, AI/ML, and Digital Twin.

Keywords*—Digital Twin, Internet of Things, Smart Building, Machine Learning, Random Forest, ESP32, Real-Time Monitoring, Occupancy Prediction, Anomaly Detection, Building Monitoring.

I. INTRODUCTION

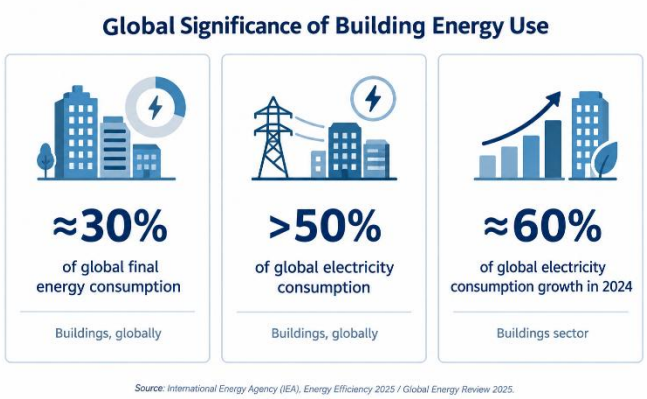
Building consist of multiple spaces that may vary in conditions throughout the day. Tracking factors such as temperature and humidity of a space may help understand activities in the space. There are numerous ways of sensing and capturing information. The Internet of Things (IoT) focuses on distributed sensing. Once sensors capture data, methods to help analyze the data are critical. Digital Twin (DT) helps bridge the gap of capturing data and representing it in a manner that is analogous to the real world. The representation helps break down, analyze, and monitor conditions of a system [1], [2]. Recent developments focused on bridging IoT and Machine learning in a enabling real time environment monitoring in buildings [3]. While progress is being made in the field, it is limited to certain spaces. Progress is made in other areas for real time building environment monitoring. Since numerous spaces in a building may require different types of sensors, methods to combine, manage and represent the captured data in a coherent format is essential for building environment monitoring.

AI and ML can analyze building data to recognize patterns and assist in condition evaluation. Several research studies have focused on AI and Digital Twins to optimize energy use in buildings and develop smart buildings [4]. Other research studies have used ML within Digital Twin frameworks to estimate indoor temperatures [5]. Other research focused on Digital Twins for real time building data visualization and building commissioning [6]. Still other research has used Digital Twins to analyze building occupants' thermal comfort and performance [7], [8]. Predicting indoor environmental quality (IEQ) of building spaces using AI and ML has also been extensively researched [9]. Integrating room-level data sensors with ML and deep learning techniques to process data and analyze environmental conditions and combining these with Digital Twin, internet of things (IoT) and other associated technologies can facilitate analysis of a broad spectrum of buildings.

An AI-based Digital Twin system is developed to monitor a prototype three-room building (Reception, Manager Cabin, Workspace) in this study. Real time measurements of surrounding environment in the prototype are acquired using an ESP32 and a array of sensors (DHT22, MQ-2 and PIR). Measurements are transmitted to a FastAPI server through Wi-Fi. Server stores the data in MongoDB. Data are analyzed after preprocessing. Room occupation prediction is performed using

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a machine learning model based on Random Forest [10]. Condition of environment in the monitored space is determined using rules. Based on rules, condition of environment is classified to normal, warning and critical state. PIR sensor is used to determine occupation of space. Digital Twin of monitored space is developed using React. Occupation and condition of monitored space is determined using rules and sensors respectively. System acquires, analyses and communicates condition of monitored space using AI and ML,



and alerts user using physical indicators. The importance of intelligent building monitoring is also reflected in the significant energy consumption of buildings, which account for about 30% of global final energy consumption and more than half of global electricity consumption , as shown in Fig. 1.

Fig. 1. Global significance of building energy use.

II. Related Works

Recent studies have been conducted on Digital Twin and IoT technologies for smart building monitoring and analysis. Boukaf et al. provided a review of Digital Twin technology for building energy consumption forecasting [11]. Hidayat et al. built a Digital Twin system that combines blockchain and mixed reality for smart buildings [12]. Tuutijärvi et al. analyzed AI-supported decision making in district heating demand response [13]. Fang studied Artificial Intelligence and Big Data in the context of smart buildings [14]. These studies combine Digital Twins, AI and IoT technologies, as well as data and analysis to improve building monitoring and operational decision-making.

Further studies have integrated AI, IoT and Digital Twin technologies for smart and sustainable building environments [15]. Other research covered Digital Twins for smart spaces beyond traditional IoT analytics [16]. Virtual Digital Twins of office spaces have also been designed using computer vision for room occupancy and energy estimation [17]. Building Digital Twins have been developed using a combination of Building Information Modeling and Computer Vision [18]. Scalable Digital Twin architectures at the edge have also been examined [19]. Web-based Building Information Modeling (BIM) platforms have been designed for building Digital Twins [20]. However, these studies generally address specific aspects such as energy analysis, visualization, occupancy estimation, BIM, computer vision, or scalable architectures. The proposed system combines IoT-based environmental sensing, Random Forest-based room occupancy prediction, rule-based environmental condition analysis, real-time backend processing, physical alerts, and room-level Digital Twin visualization within a single prototype building-monitoring system. A comparison of related systems is presented in Table I.

TABLE I. Comparison of Related Digital Twin-Based Building Monitoring Systems

Article Title	Key Objective	Technologies Used	Limitation
AI-Powered Digital Twins and Internet of Things for Smart Cities and Sustainable Building Environment	Examines the use of AI, Digital Twins, and IoT for smart-city and sustainable building applications.	Artificial Intelligence, Digital Twin, IoT, smart-building technologies, and sustainability-oriented analysis.	Provides a broad technology perspective rather than implementing a room-level monitoring prototype with physical alerts.
Digital Twins for Smart Spaces—Beyond IoT Analytics	Explores how Digital Twins can extend smart-space management beyond conventional IoT-based data analysis.	Digital Twin concepts, IoT sensing, smart-space data, and physical–virtual system integration.	Focuses on broader smart-space management and does not implement the complete room-level monitoring and alert mechanism used in this work.
VR digital twin of office space with computer vision-based estimation of room occupancy and power consumption	Develops a virtual representation of an office space for estimating occupancy and power consumption.	Virtual Reality, Digital Twin, computer vision, environmental sensing, occupancy estimation, and power-consumption analysis.	Concentrates on computer-vision-based occupancy and power estimation rather than combining multiple environmental sensors with rule-based condition analysis and physical alerts.
Computer Vision Enabled Building Digital Twin Using Building Information Model	Develops a building Digital Twin by combining Building Information Modeling with computer vision.	BIM, computer vision, building data, spatial representation, and Digital Twin technologies.	Emphasizes BIM and vision-based building representation rather than multi-parameter IoT sensing and room-level environmental condition monitoring.
Web-Based BIM Platform for Building Digital Twin	Develops a web-based BIM platform for representing and managing building Digital Twin information.	BIM, web-based visualization, building information, and Digital Twin representation.	Focuses mainly on BIM-based Digital Twin representation and does not integrate machine-learning-based occupancy prediction with multi-sensor condition analysis.

III. Methodology

The proposed system integrates IoT sensing, data communication, machine learning, rule-based condition analysis, and Digital Twin visualization for real-time building monitoring. Sensor data are collected from the physical prototype using ESP32-based sensing units and transmitted to the backend through Wi-Fi. The received data are stored, preprocessed, analysed, and presented through a room-level Digital Twin dashboard. A Random Forest machine learning

model is trained for room occupancy prediction, while rule-based analysis is used to identify abnormal environmental conditions and generate corresponding status levels.

A. System Architecture

The proposed system consists of five main layers: hardware, communication, backend, AI and analytics, and application. The hardware layer contains the ESP32 along with DHT22, MQ-2, and PIR sensors installed in the prototype rooms. The ESP32 collects the sensor readings and sends them through Wi-Fi to the FastAPI backend. The received data are stored in MongoDB and made available for preprocessing and analysis.The AI and analytics layer contains the data preprocessing stage, the Random Forest-based room occupancy model, and rule-based anomaly analysis. The machine learning model uses occupancy-related sensor features to estimate room occupancy, while the rule-based component evaluates environmental parameters and assigns normal, warning, or critical conditions. The processed information is then presented through the React-based Digital Twin dashboard, providing a connection between the physical building and its virtual representation. The complete workflow of the proposed system, from sensor data acquisition to Digital Twin visualization, is shown in Fig. 2.

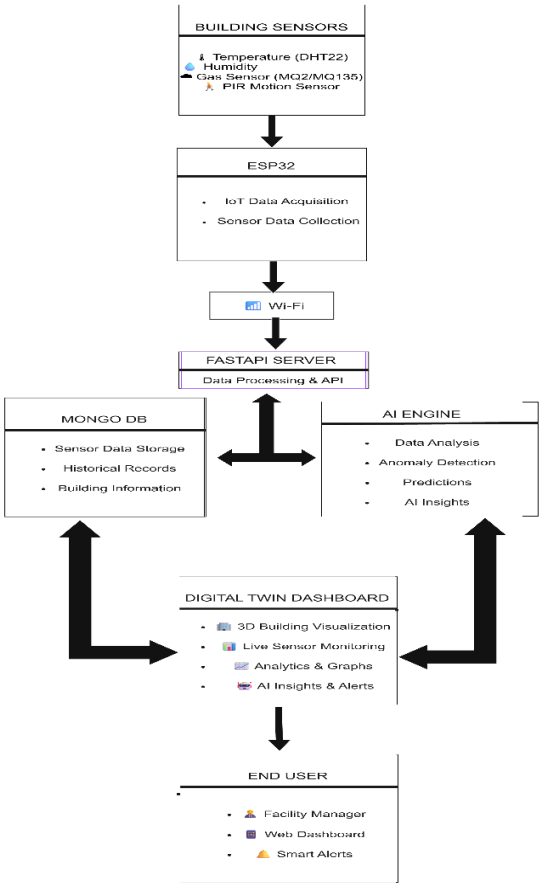


Fig. 2. Overall workflow of the proposed AI-enabled Digital Twin system.

The five-layer organization separates sensing, communication, backend processing, AI and analytics, and application-level visualization. This organization allows sensor acquisition, data processing, machine learning analysis, condition assessment, and Digital Twin visualization to operate as connected stages. The layer-wise organization of the proposed system is presented in Fig. 3.

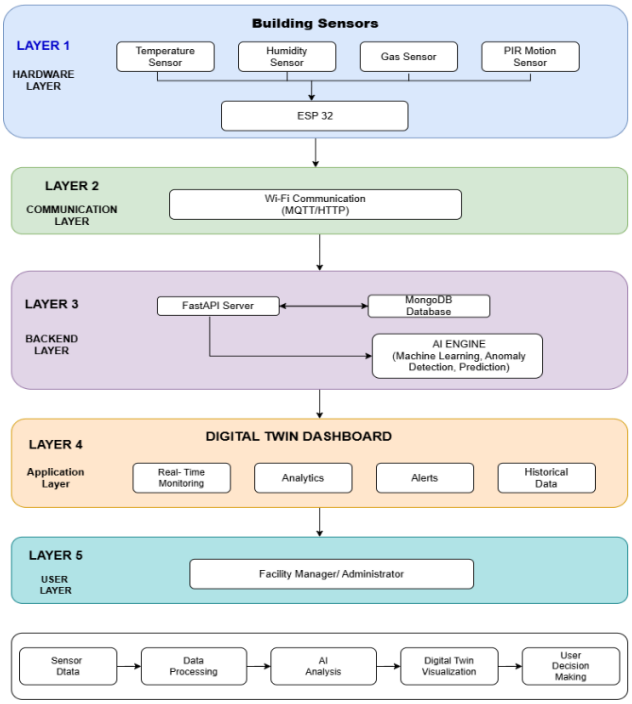


Fig. 3. Proposed five-layer architecture for AI-based Digital Twin building monitoring.

B. Sensor Data Acquisition

The physical prototype consists of three monitored spaces: Reception, Manager Cabin, and Workspace. An ESP32 is used as the main controller for collecting data from the connected sensors. DHT22 sensors provide temperature and humidity measurements, while the MQ-2 sensor is used for gas monitoring. PIR sensors detect motion and provide occupancy-related information. The sensor readings are periodically collected by the ESP32 and prepared for transmission to the backend.The implemented ESP32-based sensing setup used for real-time building data acquisition is shown in Fig. 4.



Fig. 4. ESP32-based sensor setup used for building data acquisition.

C. Data Transmission and Storage

The collected sensor readings are transmitted from the ESP32 to the FastAPI backend using Wi-Fi. The backend receives the readings and associates them with their respective rooms and sensor parameters. MongoDB is used to store the received data so that both current measurements and historical readings can be accessed by the application. This provides a continuous data flow between the physical prototype and the Digital Twin environment.The data transmission and storage process implemented in the system is shown in Fig. 5.

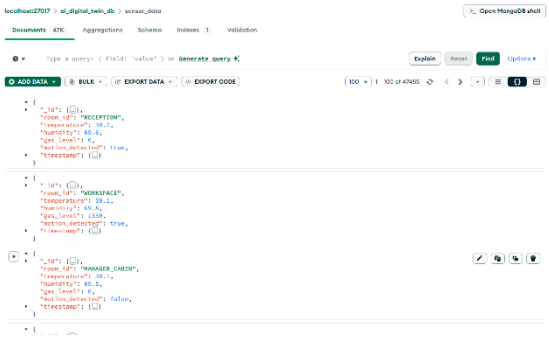


Fig. 5. Data transmission and storage process of the proposed system.

D. Data Preprocessing and AI/ML Analysis

The collected sensor data is preprocessed by extracting temperature, humidity, gas level, and motion values and normalizing them for analysis. A Random Forest classifier is trained using environmental and PIR sensor features to predict room occupancy.In parallel, a rule-based anomaly detection method classifies building conditions as normal, warning, or critical based on predefined thresholds and generates a risk score. The processed results are used by the Digital Twin dashboard for monitoring and alerts. As shown in Fig. 6.

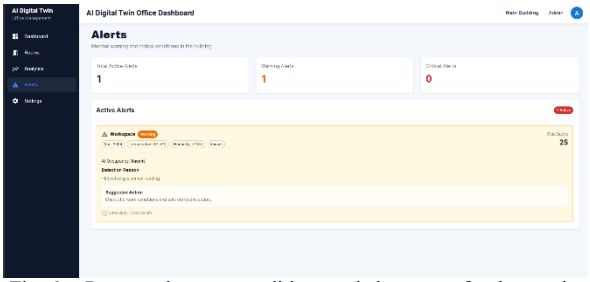


Fig. 6. Processed sensor conditions and alert states for the monitored rooms.

E. Digital Twin Dashboard

The processed information is represented in the Digital Twin dashboard as room-level information for Reception, Manager Cabin, and Workspace**.** The dashboard displays current sensor values, occupancy-related information, historical trends, status indicators, and warning conditions. This visualization allows the user to observe the condition of individual spaces through a unified interface.Physical alert mechanisms, including the buzzer and the Manager Cabin LED connected to the PIR detection system, provide local indications in addition to the software dashboard. The developed Digital Twin dashboard for real-time building monitoring is shown in Fig. 7.

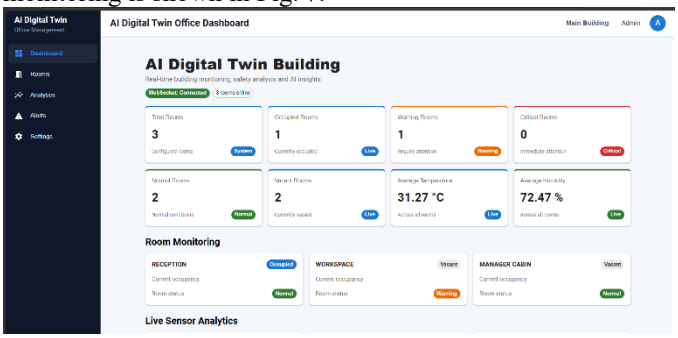


Fig. 7. Digital Twin dashboard for real-time building monitoring and visualization.

IV. Experimental Setup and Results

The system was implemented and tested using a three-room building prototype consisting of Reception, Manager Cabin, and Workspace. The setup integrates ESP32-based sensors, Wi-Fi, FastAPI, MongoDB, AI-based analysis, and a React Digital Twin dashboard. The experiments verified sensor acquisition, data transmission, storage, monitoring, abnormal-condition detection, and alerts.

A. Hardware and Software Implementation

The prototype uses an ESP32 with DHT22, MQ-2, and PIR sensors for temperature, humidity, gas, and motion monitoring. Sensor data is transmitted through Wi-Fi to FastAPI and stored in MongoDB. The React dashboard displays the processed data, while a buzzer and Manager Cabin LED provide local alerts. The hardware setup is shown in Fig. 8.



Fig. 8. Experimental hardware setup with ESP32 and connected building sensors.

B. Real-Time Sensor Monitoring

The ESP32 successfully collected temperature, humidity, gas, and motion data from all three rooms. The readings were transmitted to the backend, stored in MongoDB, and displayed on the Digital Twin dashboard for room-level monitoring. The real-time sensor values displayed by the developed system are shown in Fig. 9.

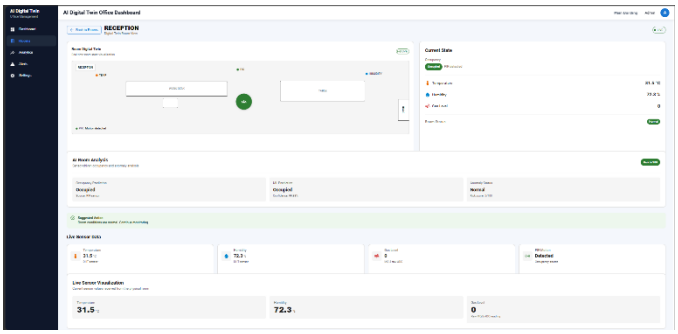


Fig. 9. Real-time room-level sensor monitoring through the Digital Twin dashboard.

C. Abnormal Condition and Alert Detection

The system was tested under different sensor conditions using threshold-based classification to identify normal, warning, and critical states. For a measured parameter, the status is determined as

$$S(x) = \begin{cases} \text{Normal}, & x < T_1 \\ \text{Warning}, & T_1 \le x < T_2 \\ \text{Critical}, & x \ge T_2 \end{cases} \qquad (1)$$

where T_1 and T_2 represent the defined warning and critical thresholds, respectively.

PIR-based occupancy-related detection was represented as

$$O = \begin{cases} 1, & \text{if motion is detected} \\ 0, & \text{otherwise} \end{cases} \qquad (2)$$

where 1 indicates detected motion and 0 indicates no detected motion. Abnormal conditions were displayed on the dashboard, while the buzzer and Manager Cabin LED provided physical alerts. The warning and alert conditions observed during testing are shown in Fig. 10.

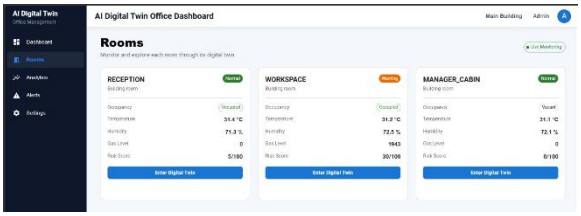


Fig. 10. Room-level Digital Twin dashboard showing environmental and occupancy conditions.

D. Data Analysis and Historical Observation

MongoDB is used to store sensor data over time. We can then go back and see historical measurements of various conditions like temperature, humidity, gas levels, and motion. We can then view these conditions in the Digital Twin dashboard to provide a room level view. By having a history of measured conditions, we can start looking for environmental condition changes and provide a basis for understanding the conditions in the monitored space. The average of a given sensor condition is calculated by adding all the readings and dividing by the total number of observations (as shown in equation 1).

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad (3)$$

where (X_{\mathrm{avg}}) is the average reading, (X_1,X_2,\ldots,X_N) represent the individual sensor readings, and (N) is the total number of observations. Environmental conditions in the Reception, Manager Cabin, and Workspace are analyzed using calculated conditions and historical data. The dashboard provides a condition comparison of monitored spaces, as well as a historical view of previously collected data for further analysis. The processed data are used to display room conditions, status indicators, and historical trends through the Digital Twin dashboard, as shown in Fig. 11.

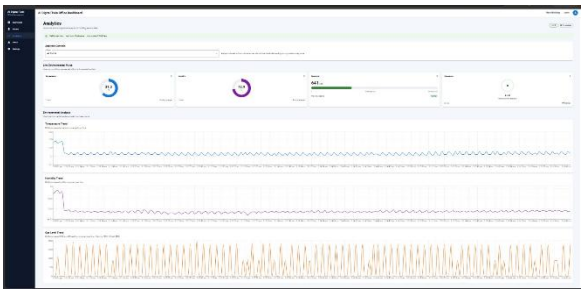


Fig. 11. Historical sensor trends and analytical information displayed in the Digital Twin dashboard.

E. Digital Twin Visualization and Results

The Digital Twin dashboard provides a virtual representation of the three monitored rooms and displays their sensor values, occupancy-related status, historical trends, and warning conditions. During testing, changes in the physical sensor conditions were reflected in the corresponding digital room status. The system also provided local alerts through the buzzer and Manager Cabin LED.

F. Experimental Observations

The experiments verified the complete data flow from physical sensing to Digital Twin visualization. ESP32

successfully acquired and transmitted sensor data, FastAPI processed the readings, MongoDB stored the data, and the dashboard displayed room-level information. Threshold testing verified the warning and critical status indications, while PIR testing confirmed motion detection and Manager Cabin LED operation. These results demonstrate the practical integration of IoT sensing, backend processing, AI-based analysis, alert mechanisms, and Digital Twin visualization in the proposed building-monitoring system. The key implementation challenges observed during development and the measures used to address them are presented in Table II.

TABLE II. IMPLEMENTATION CHALLENGES AND CORRESPONDING SOLUTIONS

Challenge Encountered	Solution Implemented
Unstable sensor readings during initial hardware testing	Sensor connections and power supply were checked, and the sensor readings were repeatedly tested to ensure stable data acquisition.
Difficulty in maintaining reliable communication between ESP32 and backend	Wi-Fi communication was configured and the ESP32 data transmission process was tested with the FastAPI backend.
Mapping sensor readings to the correct prototype rooms	Room identifiers were associated with the transmitted sensor data so that each reading could be displayed under its corresponding room.
Handling different sensor parameters in a unified system	Temperature, humidity, gas, and motion readings were structured into a common data format before backend processing and storage.
Identifying abnormal environmental conditions	Threshold-based conditions were implemented to classify readings into normal, warning, and critical states.
Synchronizing physical sensor conditions with the Digital Twin dashboard	The backend continuously processed the received sensor data and provided updated information to the React-based dashboard.
Providing physical indication for detected motion and alerts	A buzzer was integrated for alert indication, and an LED was connected to the Manager Cabin PIR sensor for motion-based indication.

V. CONCLUSION

The developed system demonstrates an AI-enabled Digital Twin for intelligent building monitoring by integrating IoT sensing, AI-based analysis, backend services, and a room-level Digital Twin dashboard. The ESP32 successfully collected temperature, humidity, gas, and motion data from the Reception, Manager Cabin, and Workspace and transmitted the readings to the FastAPI backend for MongoDB storage. The React dashboard provided real-time monitoring, historical observation, occupancy-related status, and identification of warning and critical conditions.The integration of threshold-based analysis, Random Forest-based occupancy prediction, and physical alerts through the buzzer and Manager Cabin LED connected digital monitoring with physical responses. Experimental testing verified the practical operation of the proposed prototype. The system can be further extended with additional sensors, advanced predictive models, energy analysis, predictive maintenance, and larger building deployments.

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